

Performing a Vulnerability Assessment (4e)

Fundamentals of Information Systems Security, Fourth Edition - Lab 02

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Time on Task:
2 hours, 14 minutes

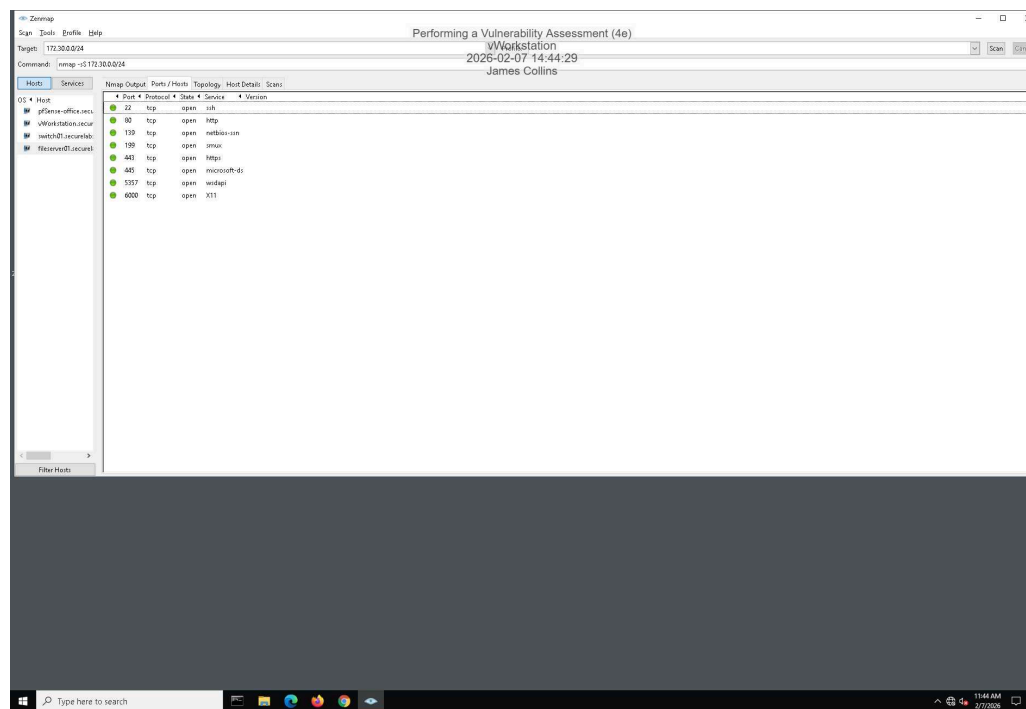
Progress:
100%

Report Generated: Sunday, February 8, 2026 at 2:14 PM

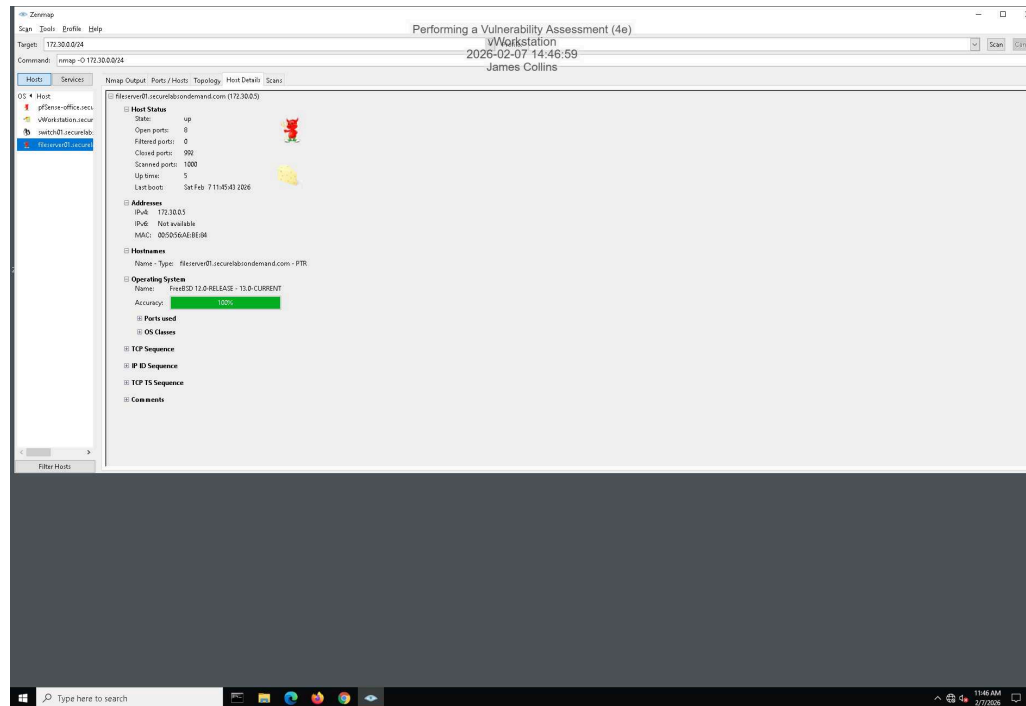
Section 1: Hands-On Demonstration

Part 1: Scan the Network with Zenmap

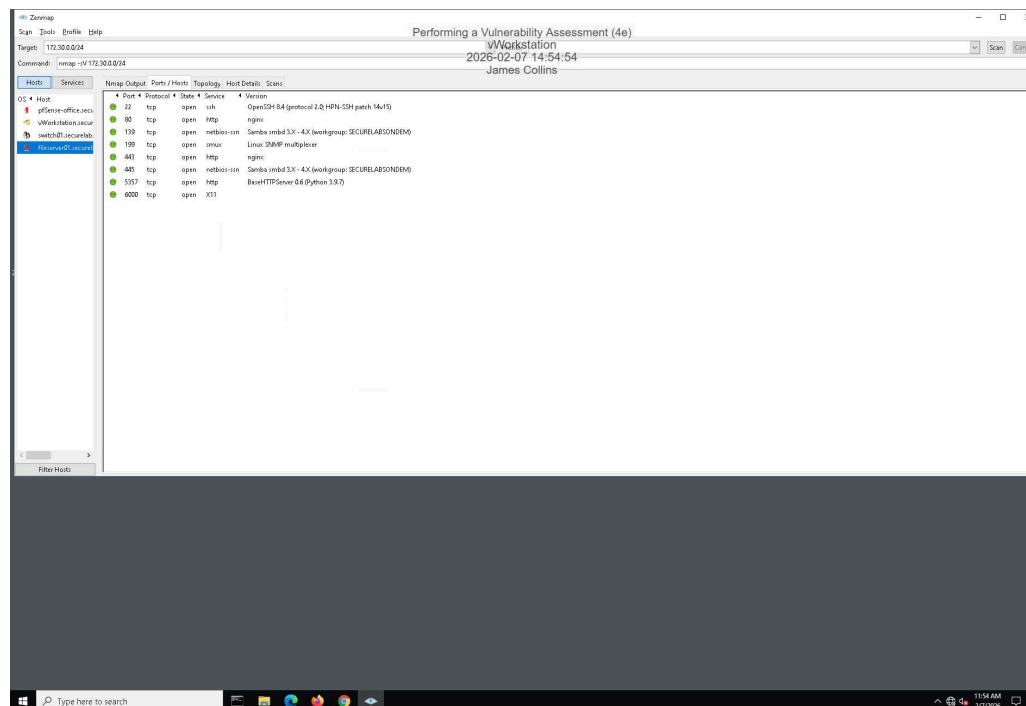
9. Make a screen capture showing the contents of the **Ports/Hosts** tab from the **SYN** scan for **fileserver01.securelabsondemand.com**.



15. Make a screen capture showing the contents of the **Host Details** tab from the OS scan for **fileserver01.securelabsondemand.com**.

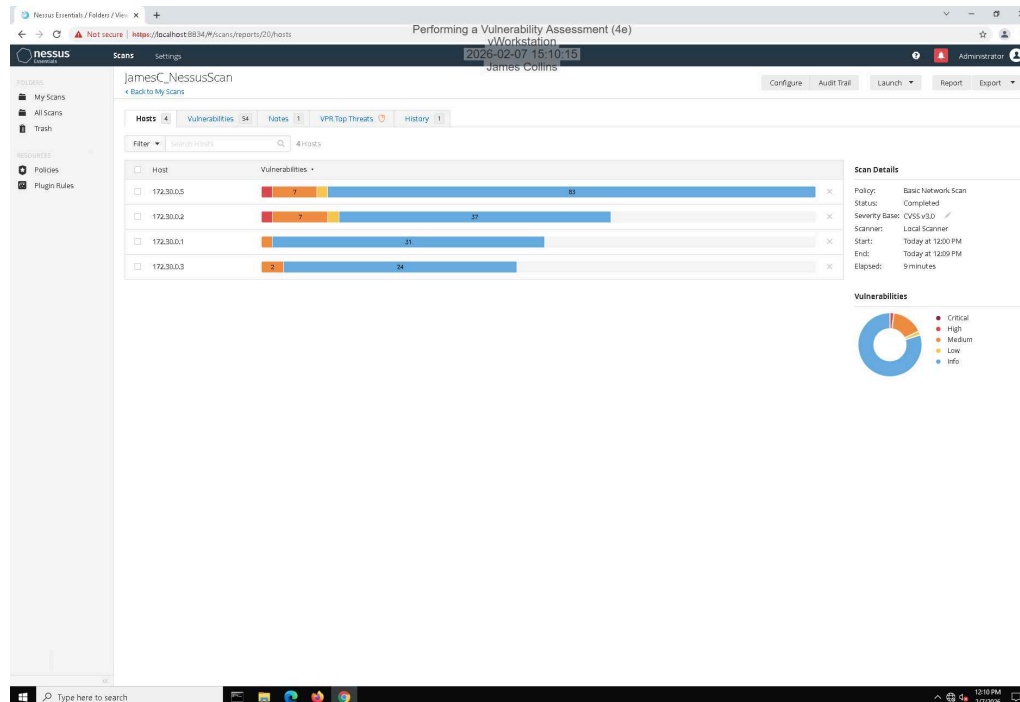


19. Make a screen capture showing the details in the **Ports/Hosts** tab from the **Service** scan for **fileserver01.securelabsondemand.com**.



Part 2: Conduct a Vulnerability Scan with Nessus

14. Make a screen capture showing the Nessus report summary.



Part 3: Evaluate Your Findings

11. **Summarize** the vulnerability you selected, including the CVSS risk score, and **recommend** a mitigation strategy.

SSL Certificate Cannot Be Trusted Summary: The scanner detected that the server is presenting a certificate that cannot be validated against a trusted Certificate Authority. This typically happens when the certificate is self-signed, the intermediate certificates are missing, the certificate is expired or not valid, or the signature cannot be verified. When the trust chain is broken, users and systems cannot reliably ensure they are communicating with a valid server.

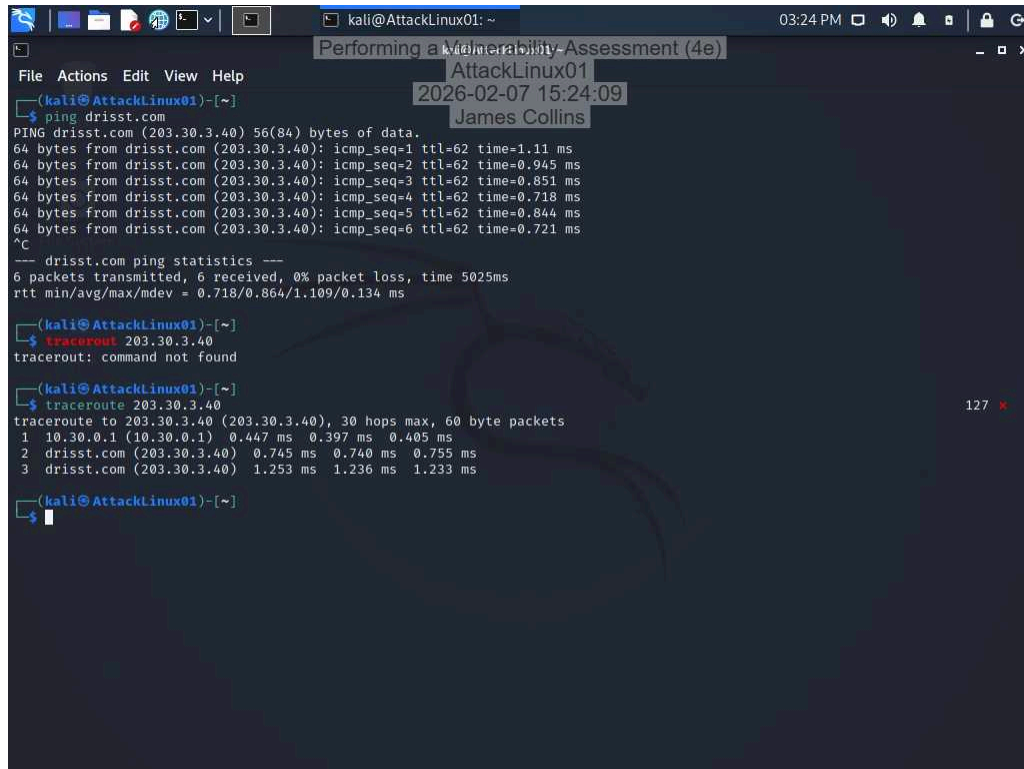
CVSS Risk Score : medium (6.5)

Mitigation Strategy: Replace the untrusted certificate with one issued by a CA and ensure the certificate is not expired and is still valid, the full certificate chain is installed, the hostname matches the certificates CN/SAN values, and then rescan to see if the issue has been fixed.

Section 2: Applied Learning

Part 1: Scan the Network with Nmap

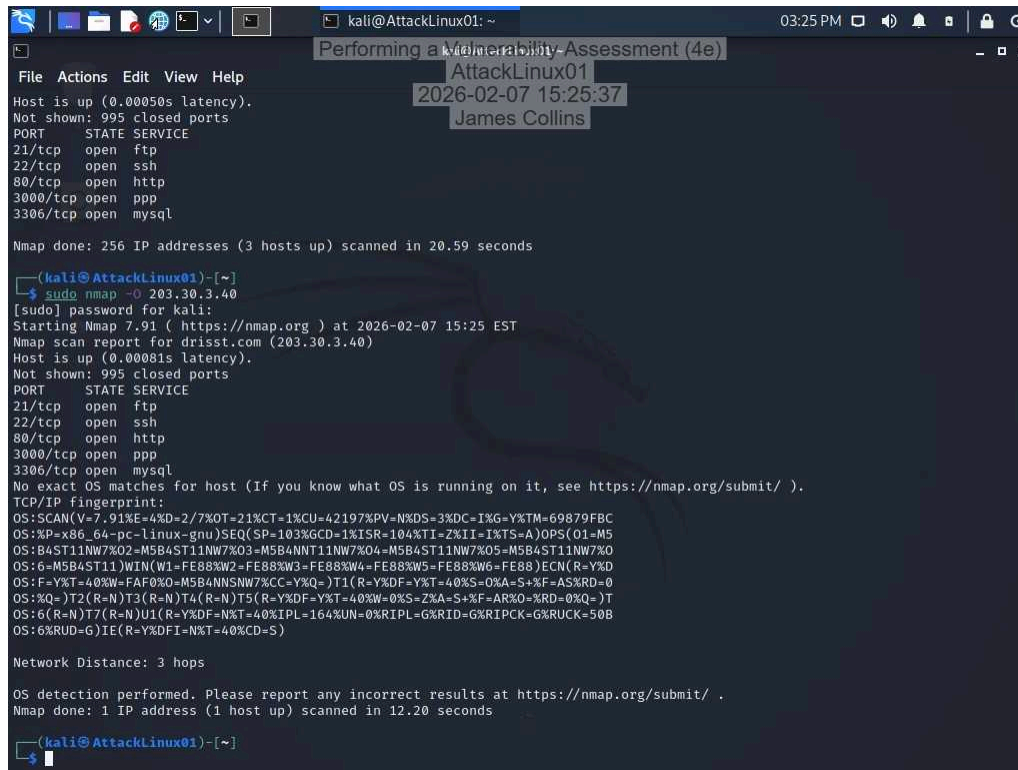
6. Make a screen capture showing the results of the traceroute command.



The screenshot shows a Kali Linux terminal window with the following content:

```
kali@AttackLinux01: ~  
Performing a Vulnerability Assessment (4e)  
AttackLinux01  
2026-02-07 15:24:09  
James Collins  
File Actions Edit View Help  
kali@AttackLinux01:~]  
$ ping drisst.com  
PING drisst.com (203.30.3.40) 56(84) bytes of data:  
64 bytes from drisst.com (203.30.3.40): icmp_seq=1 ttl=62 time=1.11 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=2 ttl=62 time=0.945 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=3 ttl=62 time=0.851 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=4 ttl=62 time=0.718 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=5 ttl=62 time=0.844 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=6 ttl=62 time=0.721 ms  
^C  
--- drisst.com ping statistics ---  
6 packets transmitted, 6 received, 0% packet loss, time 5025ms  
rtt min/avg/max/mdev = 0.718/0.864/1.109/0.134 ms  
kali@AttackLinux01:~]  
$ tracerout 203.30.3.40  
tracerout: command not found  
kali@AttackLinux01:~]  
$ traceroute 203.30.3.40  
traceroute to 203.30.3.40 (203.30.3.40), 30 hops max, 60 byte packets  
1 10.30.0.1 (10.30.0.1) 0.447 ms 0.397 ms 0.405 ms  
2 drisst.com (203.30.3.40) 0.745 ms 0.740 ms 0.755 ms  
3 drisst.com (203.30.3.40) 1.253 ms 1.236 ms 1.233 ms  
kali@AttackLinux01:~]  
$
```

10. Make a screen capture showing the results of the Nmap scan with OS detection activated.

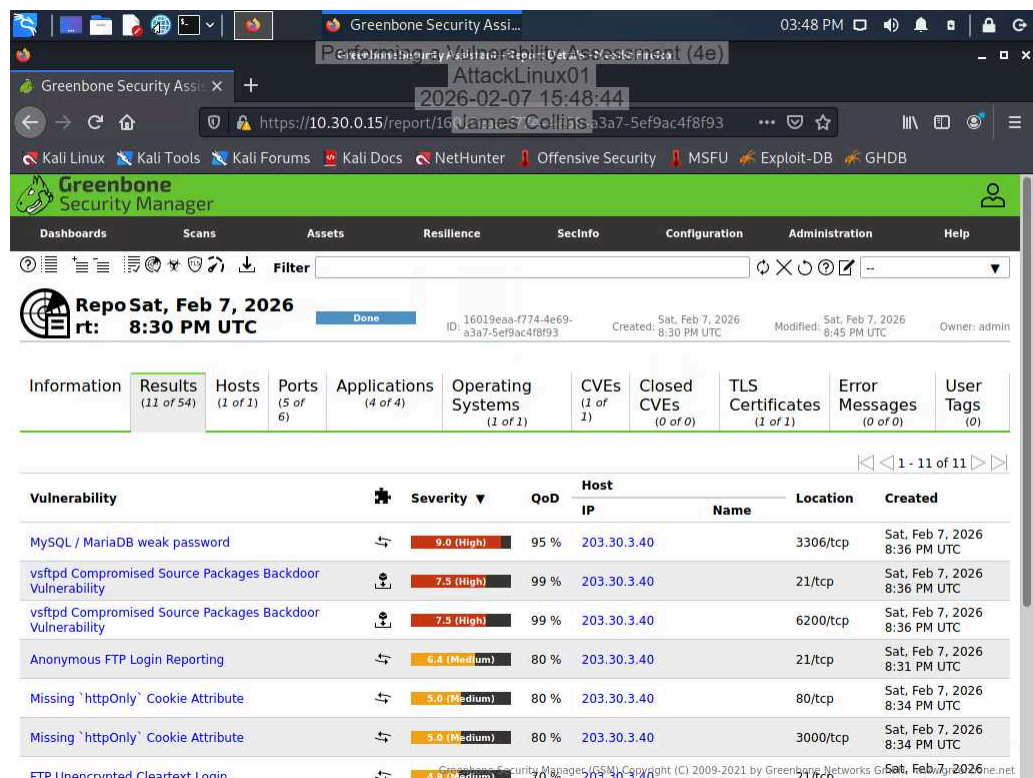


The screenshot shows a Kali Linux terminal window titled "Performing a Vulnerability Assessment (4e)". The window contains the following text:

```
kali@AttackLinux01: ~  
File Actions Edit View Help  
Host is up (0.00050s latency).  
Not shown: 995 closed ports  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
80/tcp    open  http  
3000/tcp  open  ppp  
3306/tcp  open  mysql  
  
Nmap done: 256 IP addresses (3 hosts up) scanned in 20.59 seconds  
  
kali@AttackLinux01: ~  
$ sudo nmap -O 203.30.3.40  
[sudo] password for kali:  
Starting Nmap 7.91 ( https://nmap.org ) at 2026-02-07 15:25 EST  
Nmap scan report for drisst.com (203.30.3.40)  
Host is up (0.00081s latency).  
Not shown: 995 closed ports  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
80/tcp    open  http  
3000/tcp  open  ppp  
3306/tcp  open  mysql  
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).  
TCP/IP fingerprint:  
OS:SCAN(V-7.91%E=4%D-2/7%OT-21%CT-1%CU-42197%PV-N%DS-3%DC-1%G-Y%TM-69879FBC  
OS:XP-x86_64-pc-linux-gnu)SEQ(SP-103%GCD-1%ISR-104%TI-2%II-1%TS-A)OPS(O1-M5  
OS:B4ST11NW7%O2-M5B4ST11NW7%O3-M5B4NNT11NW7%O4-M5B4ST11NW7%O5-M5B4ST11NW7%O  
OS:6-M5B4ST11WIN(W1-FE88%W2-FE88%W3-FE88%W4-FE88%W5-FE88%W6-FE88)ECN(R-Y%  
OS:F-Y%T-40%W-FAF0%O-M5B4MNSNW7%CC-Y%Q-)>T1(R-Y%DF-Y%T-40%S-0%A-S+%F-AS%RD-0  
OS:%Q-)>T2(R-N)T3(R-N)T4(R-N)T5(R-Y%DF-Y%T-40%W-0%S-Z%A-S+%F-AR%O-%RD-0%Q-)>T  
OS:6(R-N)T7(R-N)UI1(R-Y%DF-N%T-40%IPL=164UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=50B  
OS:6%RUD=G)IE(R-Y%DFI-N%T-40%CD=S)  
  
Network Distance: 3 hops  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 12.20 seconds  
  
kali@AttackLinux01: ~  
$
```

Part 2: Conduct a Vulnerability Scan with OpenVAS

13. Make a screen capture showing the detailed OpenVAS scan results.



Part 3: Prepare a Penetration Test Report

Target

Insert the target here.

Host IP: 2-3.30.3.40

Completed by

Insert your name here.

James Collins

On

Insert current date here.

2/7/2026

Purpose

Identify the purpose of the penetration test.

Evaluate security of the targeted system by identifying vulnerabilities that could potentially be used by an attacker. The assessment aimed to discover any weaknesses within the services, authentication methods, and configurations in order to determine risk and possible mitigation actions

Scope

Identify the scope of the penetration test.

The assessment focused on externally accessible services discovered during scanning. The testing included service enumeration and vulnerability identification via tools. Exploitation beyond verification of findings was not completed.

Summary of Findings

Identify and summarize each of the three high-severity vulnerabilities identified during your penetration test. For each vulnerability, identify the severity, describe the issue, and recommend a remediation.

The most critical issue was the MYSQL/MariaDB database allowed a successful login as root using a very weak password, which could let an attacker access or change sensitive data. the FTP service was also running vsftpd 2.3.4, a version known to include a backdoor port (6200) was reachable, increasing the chance of a full system compromise. Fixing these problems will include changing the weak passwords, limiting admin access, reinstall the FTP service from a trusted source, and block any unneeded ports.

Conclusion

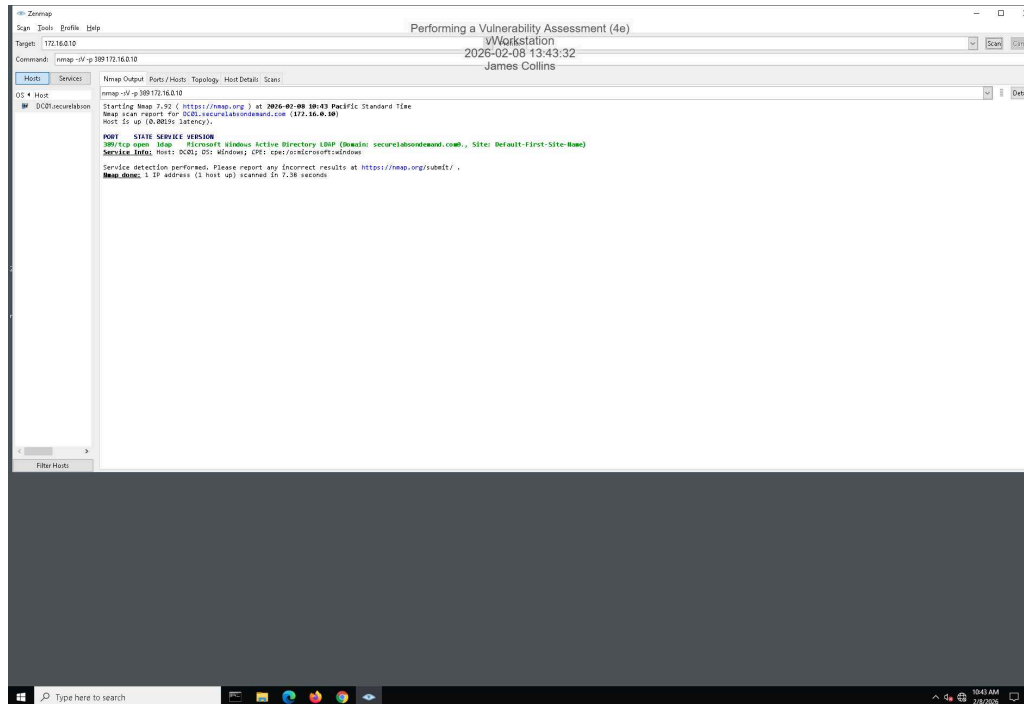
Identify your key findings.

The vulnerability scan identified multiple serious security issues, including weak authentication controls and a known backdoored FTP service. The most critical vulnerability found was the ability to authenticate to the database as root using a common password. Also, the presence of a compromised vsftpd package raises the risk of RCE. These findings show that this system is at a severely high risk of compromise, if exposed to untrusted networks. Immediate remediation strategies must be done to reduce the attack surface.

Section 3: Challenge and Analysis

Part 1: Scan the Domain Controller with Nmap

Make screen capture showing the results of your targeted port scan on the domain controller.

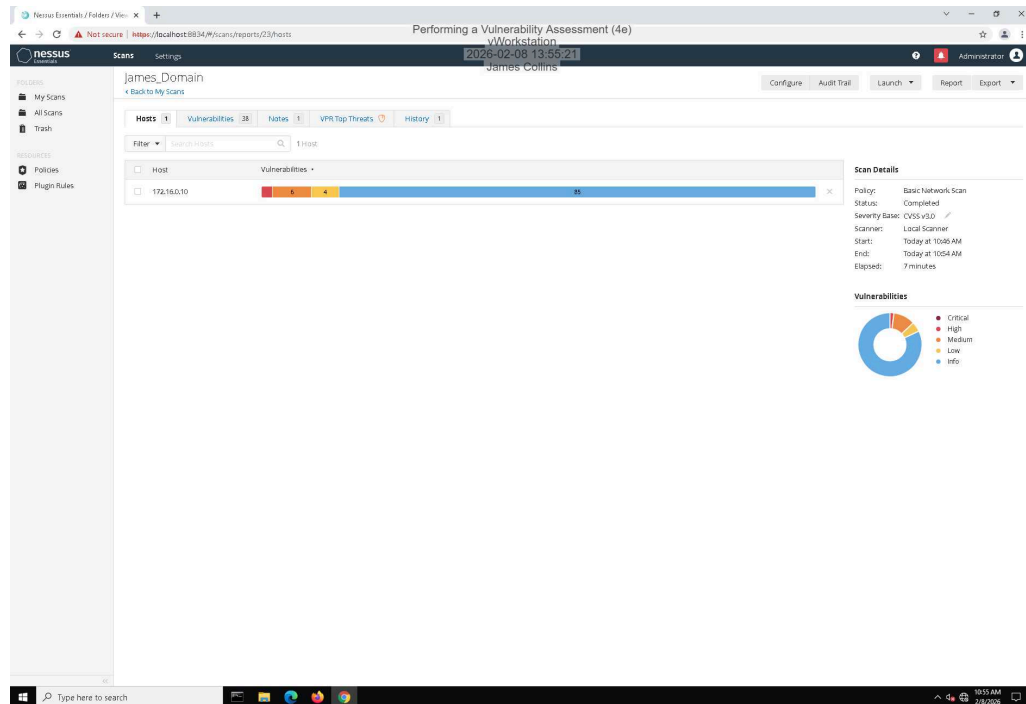


Part 2: Scan the Domain Controller with Nessus

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Make a screen capture showing the Nessus report summary for the domain controller.



Part 3: Prepare a Penetration Test Report

Target

Insert the target here.

172.16.0.10

Completed by

Insert your name here.

James Collins

On

Insert current date here.

2/8/2026

Purpose

Identify the purpose of the penetration test.

TO identify security weaknesses on the target system and evaluate the potential risks.

Scope

Identify the scope of the penetration test.

Scope was limited to 172.16.0.10 and testing was not intrusive and focused on configuration and protocol weaknesses and not exploitation.

Summary of Findings

Identify and summarize each vulnerability identified during your penetration test. For each vulnerability, identify the severity, describe the issue, and recommend a remediation.

SSL Medium Strength Cipher Suites SupportedSeverity is HighPlugin id 42873CVE-2016-2183Score of 7.5

The host supports ssl/tls cipher suites considered medium strength and in this case the service allows the use of triple DES. Due to the 64 bit block sized used by triple DES, long lived encrypted sessions are vulnerable to a birthday attack called SWEET32. AN attacker may be able to recover portions of plaintext.

Successful exploitation would allow an attacker to decrypt sensitive information transmitted over the connection.

Reconfigure the affected service to disable the triple DES and any other medium-strength ciphers. Only strong, modern ciphers should be permitted. Perform a scan once changes are completed to ensure the weak suites are no longer being allowed.

Conclusion

Identify your key findings.

The scan identified a high-severity cryptographic configuration weakness on the target host. Support for triple DES makes encrypted sessions vulnerable to SWEET32 birthday attack and puts sensitive communications at risk. The vulnerability should be remediated by enforcing stronger encryption standards.